Module 1 knowledge check

Which Service type should you use when you need to expose an application to the internet with an Azure-managed load balancer?

A. LoadBalancer
B. ClusterIP
C. NodePort

What must match between a Deployment and a Service for traffic to route correctly?

A. The Pod labels in the Deployment template must match the Service selector
B. The container port in the Deployment must match the Service port
C. The Deployment name must match the Service name

Module 2 knowledge check

You need to store a database connection string that contains a password. The value shouldn't be visible in source control. Which Kubernetes resource should you use?

A. ConfigMap, because it stores configuration data
B. Secret, because it stores sensitive values and keeps credentials out of source control
C. PersistentVolumeClaim, because it provides storage for application data

Your application reads a JSON configuration file at startup. You want to update the file without rebuilding the container image. How should you provide the configuration?

A. Mount a ConfigMap as files using a volume so the JSON file appears on disk
B. Store the JSON in a Secret and inject it as an environment variable
C. Embed the JSON in the container image and rebuild when it changes

Module 3 knowledge check

An AI inference API returns intermittent HTTP 500 errors. You want to inspect error messages for a specific pod while reproducing the issue. Which approach is most appropriate?

A. Use kubectl logs -f to stream logs from the pod while sending test requests
B. Open Azure Monitor and review node CPU metrics for the last week
C. Run kubectl describe node on all nodes to check scheduling events

A Service shows no endpoints even though pods appear healthy. Which command confirms whether Service selectors match pod labels?

A. kubectl describe service
B. kubectl top nodes
C. kubectl logs